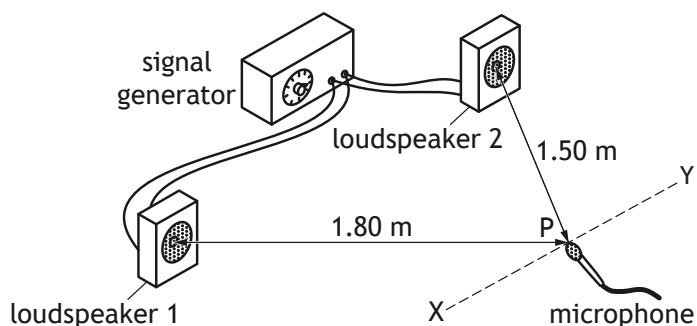


10. A group of students is investigating interference of sound waves.
The students connect two loudspeakers to the same output of a signal generator.



The signal generator produces a signal of frequency 1700 Hz.

- (a) Calculate the wavelength of the sound produced by the speakers.

3

Space for working and answer

- (b) As the microphone is moved from X to Y, regions of maxima and minima are detected.

Point P is a distance of 1.80 m from loudspeaker 1 and a distance of 1.50 m from loudspeaker 2.

Show by calculation whether constructive interference or destructive interference is detected at point P.

4

Space for working and answer



10. (continued)

- (c) Loudspeaker 1 is now disconnected from the signal generator.

State the effect this has on the amplitude of the sound detected by the microphone at point P.

Justify your answer.

2

[Turn over



* X 8 5 7 7 6 0 1 3 5 *